Technical Specifications and Platform Overview

Secuura Verification System (SVS)

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0.9.0	17-02-2026	Secuura Engineering	Platform hardening, Outlook Add-in, port migration
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1. Executive Summary

> **Secuura** is a blockchain-anchored document authenticity platform built on the **Secuura Verification System (SVS)**. It enables stakeholders to certify, verify, and manage the provenance of documents and credentials without Secuura itself acting as a certifier or verifier. For a full description of the platform's purpose and boundaries, see Section 2 — Platform Vision and Intent.

The platform operates on a fundamental principle: **Secuura enables and facilitates verification — it does not certify or verify itself.** Documents never leave the stakeholder's device; only cryptographic hashes are transmitted and anchored on the Cardano blockchain, providing tamper-evident, immutable proof of authenticity. The hashing approach is detailed in Section 8.2 — Encryption and Hashing.

21 Backend Microservices	**6** Frontend Applications
50+ Database Tables	**W3C** VC Data Model 1.1

Key Capabilities

- **Client-side document hashing** using Web Crypto API (SHA-256 / BLAKE2b-256) — see Section 8.2
- **Blockchain anchoring** on Cardano (single and Merkle-batch via metadata labels 674/675) — see Section 6.4
- **W3C Verifiable Credentials** with Ed25519 signatures and StatusList2021 revocation — see Section 6.5
- **Decentralised Identity (DID)** via did:prism and did:key methods — see Section 9.3
- **Zero-knowledge proofs** for selective disclosure via Midnight protocol — see Section 6.9
- **DAO governance** with token-weighted voting and on-chain proposal execution — see Section 10
- **Microsoft 365 integration** including SharePoint, Entra ID SSO, and Outlook Add-in — see Section 7.1 and Section 7.2
- **NFT certificate minting** with multi-tier storage (on-chain, IPFS, Arweave) — see Section 7.4
- **GDPR-compliant** data subject request handling, consent management, and right to erasure — see Section 9.1
- **RFC-3161 timestamping** via eIDAS-compliant TSA integration — see Section 9.5

Technology Stack

Layer	Technologies	Details
Frontend	React 18, TypeScript, Material UI, Vite, Office.js	Section 5.3
Backend	Node.js, Express, TypeScript	Section 5.2
Database	PostgreSQL 16, Redis 7	Section 5.5
Blockchain	Cardano (Blockfrost, CIP-30/CIP-8), Midnight	Section 7.3
Identity	JWT, bcrypt, TOTP MFA, CIP-30, Entra ID	Section 6.3
Infrastructure	Docker, Nginx, Prometheus, Grafana	Section 5.4
Standards	W3C VC 1.1, DID Core, GDPR, NIST CSF, eIDAS	Section 9

2. Platform Vision and Intent

The Secuura Verification System (SVS) addresses a fundamental trust challenge: How can a party verify that a document or credential is authentic, without relying on a single centralised authority? The complete stakeholder model that answers this question is described in Section 3 — Stakeholders and Personas.

2.1 What SVS Is

- A platform for **Verifiers to verify** that a Certifier has certified information
- An **enabler and facilitator** for Stakeholders to manage document authenticity
- A **permissionless rights management system** where Stakeholders control their own Rights (see Section 4.4)
- A **provenance tracking system** that records the complete lifecycle of certifications (see Section 4.3)
- A **blockchain-anchored proof system** providing tamper-evident, immutable records (see Section 6.4)

2.2 What SVS Is Not

- **Not a Certifier or Verifier** — Secuura does not certify or verify documents itself (see Certifier and Verifier roles)
- **Not a file storage service** — original documents are never stored on Secuura servers (see Core Design Principles)
- **Not an encryption/data protection solution** — Secuura anchors hashes, not encrypted content
- **Not an e-signing service** — it is not a DocuSign competitor
- **Not a workflow/BPM tool** — it facilitates certification and verification, not business processes

2.3 Core Design Principles

Principle	Implementation	Reference
Client-Side Hashing	All document hashing is performed in the browser using Web Crypto API. Documents never leave the user's device.	Section 8.2
No Document Storage	Only hashes, metadata, and URIs are stored. Original documents remain with Stakeholders.	Section 5.5
Permissionless Rights	Stakeholders control their own Rights without central authority approval.	Section 4.4
Immutable Provenance	All certification events form a linked chain anchored on the Cardano blockchain.	Section 4.3
Standards-Based	W3C Verifiable Credentials, DID Core, OpenAPI 3.0, and eIDAS compliance.	Section 9
Privacy by Design	GDPR-compliant from inception with zero-knowledge proof support for selective disclosure.	Section 9.1, Section 6.9

3. Stakeholders and Personas

The SVS defines five core stakeholder roles that interact with the platform. Each role has distinct capabilities, permissions, and user flows. A single user may hold multiple roles simultaneously. For the complete certification and verification flows involving these stakeholders, see Section 4 — User Flows and Certification Lifecycle.

3.1 Issuer

Definition

The Issuer is the entity that **creates the information or document**. This could be a university issuing a degree, a government agency issuing a licence, or a corporation issuing a certificate of employment.

Capabilities

- Create and register documents on the platform (see Document Certification Flow)
- Generate client-side document hashes (see Section 8.2)
- Nominate authorised signatories within their organisation
- Initiate the certification process
- View document lineage and provenance (see Certification Phases)

Portal Access

Issuer Portal (`/`) — Full document management, upload, certification initiation, wallet connection, privacy controls, and settings including Outlook Add-in setup. See Section 5.3 for frontend details.

3.2 Certifier

Definition

The Certifier **attests to the authenticity and integrity** of information. The Certifier may be the same entity as the Issuer, or a separate trusted party. Certification creates an on-chain record.

Capabilities

- Review pending certification requests
- Digitally sign documents using CIP-30 wallet (see Cardano Blockchain)
- Anchor certification hashes on the Cardano blockchain (see Blockchain Services)
- Revoke certifications when necessary
- Share certifications with other stakeholders

Certification Process

The Certifier attests that a document is authentic by signing a certification record. This record includes the document hash, the Certifier's DID, a timestamp, and is anchored on the Cardano blockchain via metadata label 674 (single) or 675 (Merkle batch). See Section 4.1 for the full flow and Section 9.4 for Cardano best practices.

3.3 Subject

Definition

The Subject is the **person or entity that the information pertains to**. For example, the graduate named on a degree certificate, or the employee referenced in a work reference.

Capabilities

- View certifications about themselves
- Control privacy preferences for their data
- Request selective disclosure via zero-knowledge proofs (see Section 6.9)
- Exercise GDPR data subject rights — access, rectification, erasure, portability (see Section 9.1)

3.4 Rights Holder

Definition

The Rights Holder has **ownership or control rights over the information**. They can share, transfer, grant access, and revoke access to certifications. See Section 4.4 — Rights Management for the full rights model.

Supported Rights

Right	Description
`CERTIFY`	Attest to the authenticity of information
`ACCESS`	View the information and its certification
`SHARE`	Provide access to other stakeholders
`TRANSFER`	Move ownership or rights to another party
`VERIFY`	Check the certification status
`VIEW_PROVENANCE`	See the complete history of the document
`MODIFY_RIGHTS`	Change rights granted to others
`REVOKE_ACCESS`	Remove previously granted access

3.5 Verifier

Definition

The Verifier **checks that a Certifier has certified the information**. Verification does not verify the content of the document itself — it confirms that a trusted Certifier has attested to its authenticity. This distinction is central to the SVS model (see Section 2.1).

Capabilities

- Verify certifications by document ID, file hash, or QR code (see Verification Flow)
- View certification details (Certifier, date, blockchain anchor)
- Record verification events on the platform (a chargeable event)
- Initiate a new certification phase by adding a verification note

Portal Access

Verifier Portal (`/verify/`) — Public verification interface supporting verification by ID, file upload (client-side hashing), and QR code scanning. See Section 5.3 for frontend details.

3.6 System Administrator

Definition

The System Administrator manages platform configuration, user accounts, organisations, integrations, and system health.

Capabilities

- Manage users, organisations, and roles (see Authentication Service)
- Configure document types and workflows
- Manage API keys and third-party verifiers
- Monitor system health across all services (see Infrastructure)
- Configure integrations — M365, blockchain, KYC providers, Office Add-ins
- View audit logs and generate reports (see Analytics)
- Process GDPR data subject requests (see GDPR Compliance)

Portal Access

Admin Dashboard (`/admin/`) — Full platform administration including Integration Hub with Office Add-in management. See Section 5.3.

4. User Flows and Certification Lifecycle

This section describes the core workflows that Stakeholders follow when interacting with the platform. Each flow involves multiple backend services working together through the API Gateway.

4.1 Document Certification Flow

1. **Document Creation** — Issuer creates or selects a document in the Issuer Portal (or via the Outlook Add-in).
2. **Client-Side Hashing** — The document is hashed locally in the browser using `crypto.subtle.digest('SHA-256', buffer)`. The file never leaves the device. See Section 8.2.
3. **Hash Registration** — Only the hash, title, and metadata are sent to the Originate Service via the API Gateway.
4. **Certification Signing** — An authorised Certifier signs the certification record using a CIP-30 wallet (see Cardano Blockchain).
5. **Blockchain Anchoring** — The Anchoring Service anchors the hash on the Cardano blockchain (metadata label 674 for single, 675 for Merkle batch).
6. **VC Issuance** — The VC Issuer Service creates a W3C Verifiable Credential with Ed25519 proof and StatusList2021 status.
7. **Timestamping** — An RFC-3161 timestamp is obtained from an eIDAS-compliant TSA for legal non-repudiation. See Section 9.5.
8. **Billing Event** — A chargeable event is recorded (certification signing + blockchain anchor).

4.2 Verification Flow

1. **Verification Request** — Verifier accesses the Verifier Portal (Section 5.3) and selects a verification method (ID, file upload, or QR scan).
2. **Hash Comparison** — If verifying by file, the document is hashed client-side (Section 8.2) and compared against the stored hash.
3. **Certification Check** — The Originate Service verifies that a valid certification exists for the document hash.
4. **Blockchain Verification** — The blockchain anchor is verified via the Blockfrost API to confirm the transaction exists (see Section 7.3).
5. **VC Verification** — The Verifiable Credential is verified (structure, signature, status, expiration, issuer DID, blockchain anchor).
6. **Result Display** — The verification result is presented with Certifier details, certification date, blockchain transaction, and provenance chain.

4.3 Certification Phases

Certifications progress through phases. A Verifier can become a Certifier by adding a verification note, which creates a new linked phase. This mechanism is central to the SVS model described in Section 2.1.

Phase	Actor	Action	Blockchain Record
Phase 1 Start	Certifier	Certifies document	Certification hash anchored (label 674/675)
Phase 1 Active	Any Stakeholder	Shares, grants access	Access events logged
Phase 1 End	Verifier	Verifies certification	Verification event recorded
Phase 2 Start	Verifier (now Certifier)	Adds note, creates new phase	New phase hash anchored, linked to Phase 1

Phases are linked via `previousPhaseId` and all remain accessible in perpetuity on the blockchain. The database stores these in the `certification_events` table (see Appendix B).

4.4 Rights Management

The SVS implements a permissionless rights model. Stakeholders control their own Rights without requiring platform approval. The Rights Holder persona has full details on the supported rights.

```
enum Right {
  CERTIFY, ACCESS, SHARE, TRANSFER,
  VERIFY, VIEW_PROVENANCE, MODIFY_RIGHTS, REVOKE_ACCESS
}

interface RightsGrant {
  stakeholderId: string;
  rights: Right[];
  grantedBy: string;
  grantedAt: string;
  expiresAt?: string;
}
```

Each rights change is a chargeable event (specifically `TRANSFER_RIGHTS` and `REVOKE_ACCESS`).

5. Solution Architecture

This section provides a comprehensive view of the platform's technical architecture. For individual component specifications, see Section 6. For external integration details, see Section 7.

5.1 High-Level Architecture

Secuura employs a microservices architecture with a unified API Gateway, reverse-proxy gateway, and dedicated frontend applications for each stakeholder role.

Architecture Layers

Layer	Components	Technology	Details
Presentation	Issuer, Verifier, Admin, Status, Website, Outlook Add-in	React, TypeScript, MUI, Office.js	Section 5.3
Gateway	Nginx Reverse Proxy, API Gateway	Nginx 1.25, Express	Section 6.1
Application	21 microservices	Node.js, Express, TypeScript	Section 5.2
Data	PostgreSQL, Redis, PGBouncer	PostgreSQL 16, Redis 7	Section 5.5
Blockchain	Cardano node, Blockfrost API, Midnight	Cardano Preprod/Mainnet	Section 7.3
External	Microsoft Graph API, Stripe, IPFS, KYC, TSA	REST APIs	Section 7

5.2 Backend Microservices

All services follow a consistent pattern: Express server, PostgreSQL connection with in-memory fallback, Redis for caching/sessions, health endpoint, and structured Winston logging. Complete port mapping is available in Appendix A.

Service	Internal Port	External Port	Purpose	Status
API Gateway	8080	—	Request routing, auth, rate limiting, Swagger docs	Active
Originate	4000	6000	Document certification, verification, GDPR, provenance	Active
Auth	4003	6003	Authentication, JWT, MFA, wallet auth, sessions	Active
PRISM	4001	6001	DID management, credential issuance	Active
Anchoring	4005	6005	Blockchain anchoring (single and Merkle batch)	Active
Wallet Connector	4002	6002	CIP-30 wallet integration	Active
Timestamping	4004	6004	RFC-3161 timestamping	Active
Security	4008	6008	Audit logs, API key management, security events	Active
Analytics	4010	6010	Platform metrics, reports, real-time analytics	Active
M365 Integration	4013	6013	SharePoint, Entra ID, Outlook Add-in support	Active
VC Issuer	4014	6014	W3C Verifiable Credential issuance, StatusList2021	Active
Staking	4015	6015	Token staking, tier management, rewards	Active
Referral	4016	6016	Referral programme, milestones, leaderboard	Active
Governance	4017	6017	DAO governance, proposals, voting	Active
NFT Certificate	4018	6018	NFT minting, multi-tier storage, IPFS	Active
Billing	4019	6019	Stripe payments, certification bundles, chargeable events	Active
Transfer	4020	6020	Ownership transfers, delegations	Active
KYC	4011	6011	Identity verification (Onfido, Jumio)	Active
Midnight	4012	6012	Zero-knowledge proofs, selective disclosure	Active
Guardian	4006	6006	Guardian recovery (planned)	Phase 2
Queue	4007	6007	Message queue (planned)	Phase 2

5.3 Frontend Applications

Each frontend is a React 18 single-page application built with Vite and served as static assets via the Nginx gateway. For the stakeholder roles served by each portal, see Section 3.

Application	Dev Port	Gateway Route	Technology	Purpose
Issuer Portal	6100	`/`	React, MUI, Vite	Document management, certification, wallet, settings
Verifier Portal	6101	`/verify/`	React, Vite	Public verification by ID, file, QR
Admin Dashboard	6102	`/admin/`	React, MUI, Recharts, Vite	Platform administration, integration hub
Website	6103	`/website/`	React, Vite	Marketing, tokenomics, whitepaper
Status Page	—	`/status/`	React	Public system health status
Outlook Add-in	6104	—	React, Office.js, Fluent UI	Verify/certify from Outlook email

5.4 Infrastructure and Deployment

Container Orchestration

All services are containerised with Docker and orchestrated via Docker Compose. Each service has a dedicated Dockerfile with multi-stage builds for production optimisation. Security controls for containers are described in Section 8.4.

Reverse Proxy (Nginx Gateway)

A single Nginx instance on port **6882** serves as the unified entry point, routing requests to all frontend applications and backend services. It provides security headers, WebSocket support (HMR), and production-ready TLS configuration. See Appendix A for the complete port reference.

Database

PostgreSQL 16 with SCRAM-SHA-256 authentication, initialised via SQL migration scripts in `docker/init/`. PGBouncer provides connection pooling in production. Redis 7 provides session storage, rate limiting, and caching with AOF persistence and 256MB memory limit. See Section 5.5 for schema details.

Monitoring

Prometheus metrics collection and Grafana dashboards are available via the `monitoring` Docker profile on ports 6301 and 6300 respectively (see Appendix A).

5.5 Database Architecture

The PostgreSQL database contains 50+ tables organised across functional domains. A full table reference is provided in Appendix B.

Domain	Key Tables	Purpose	Related Section
Users & Auth	`users`, `user_sessions`, `refresh_tokens`, `wallet_registrations`	Identity, authentication, sessions	Section 6.3
Organisations	`organizations`, `organization_members`	Multi-tenant organisation model	—
Documents	`documents`, `document_versions`, `document_signatures`, `document_access`	Document records, versioning, access control	Section 6.2
Blockchain	`blockchain_anchors`, `merkle_batches`	Cardano transaction records	Section 6.4
Provenance	`certification_events`	Linked chain of all certification lifecycle events	Section 4.3
Identity	`did_documents`, `verifiable_credentials`	DIDs and VCs	Section 6.5
GDPR	`consent_records`, `data_subject_requests`, `data_deletion_log`	Compliance records	Section 9.1
Governance	`svc_proposals`, `svc_governance_participants`, `svc_vote_delegations`	DAO voting and proposals	Section 10
Billing	`billing_charge_events`, `pricing_config`	Revenue tracking, pricing	Section 6.7
Privacy	`svc_zk_proofs`, `svc_commitments`, `svc_disclosure_audit`	Zero-knowledge proofs	Section 6.9
Admin	`document_type_configs`, `workflow_configs`, `admin_integrations`	Platform configuration	—

5.6 Network and Port Configuration

All services communicate via an internal Docker bridge network (`secuura-network`). Only two ports are exposed externally. The full port reference is in Appendix A.

> **Port 6882** — Nginx Gateway (unified entry point for all portals and APIs)
>
> **Port 6881** — Marketing Website (direct access)

Internal service ports use the 4000–4020 range. Host-mapped debug ports use the 6000–6020 range. Frontend dev servers use the 6100–6104 range. Infrastructure services use the 6300–6432 range.

6. Component Specifications

This section provides detailed specifications for each major platform component. For the overall architecture, see Section 5. For external integrations, see Section 7. For the complete API endpoint catalogue, see Appendix C.

6.1 API Gateway

The API Gateway serves as the central entry point for all API requests, providing authentication, rate limiting, input sanitisation, and request routing to backend services. For full security details, see Section 8.3.

Feature	Implementation	Reference
Rate Limiting (Auth)	5 login attempts per 15 minutes (production), 10 (development)	Section 8.3
Rate Limiting (API)	60 requests/minute (production), 100 (development)	Appendix D
Rate Limiting (Verification)	100 requests/minute (production), 200 (development)	—
Brute Force Protection	5 free retries, exponential backoff (500ms–30s), 5-minute lockout	Section 8.3
Input Sanitisation	HTML entity encoding for XSS prevention on all request bodies	Section 8.3
Suspicious Request Detection	Pattern matching for path traversal, XSS, SQL injection, command injection	NIST: Detect
JWT Validation	HS256 with configurable secret; production fails if default secret is used	Section 8.1
CSRF Protection	Token-based CSRF middleware with configurable secret	Section 8.3
Security Headers	Helmet.js with HSTS, X-Frame-Options (DENY), CSP, Permissions-Policy	Section 8.3
API Documentation	Swagger UI at `/api/docs`, OpenAPI 3.0 spec at `/api/docs/openapi.json`	—

6.2 Originate Service

The core document management service handling certification, verification, GDPR compliance, provenance tracking, and administrative configuration. This is the primary service behind the Document Certification Flow and Verification Flow.

Key Endpoint Groups

- `/api/certifications/*` — Issue, revoke, verify, share, recertify, lineage, evidence bundles (see Section 4.1)
- `/api/documents/*` — Create, sign, anchor, QR generation
- `/api/verification/*` — Hash verification, integrity checks (see Section 4.2)
- `/api/gdpr/*` — Consent management, DSR handling, data export, erasure, retention (see Section 9.1)
- `/api/signatories/*` — Authorised signatory management
- `/api/third-party-verifiers/*` — Third-party verifier registration and management
- `/api/admin/*` — Pricing configuration, document types, workflows
- `/api/system-errors/*` — Error tracking and ingestion

Full endpoint listing in Appendix C.

6.3 Authentication Service

Authentication Methods

Method	Mechanism	Use Case	Reference
Email / Password	bcrypt hashing, JWT access + refresh tokens	Standard user authentication	Section 8.1
Wallet (CIP-30)	CIP-8 challenge/response, wallet signature verification	Cardano wallet holders	Section 7.3, Section 9.4
Microsoft Entra ID	OAuth 2.0 via MSAL, SSO token exchange	Enterprise users with M365	Section 7.1
MFA (TOTP)	Time-based one-time passwords with backup codes	Enhanced account security	Section 8.1

Token Management

- **Access Tokens:** Short-lived JWT stored in-memory (not localStorage) — see Section 8.1
- **Refresh Tokens:** httpOnly cookies with automatic rotation
- **Sessions:** Redis-backed with configurable expiry and revocation support

6.4 Blockchain Services

Anchoring Service

Anchors document hashes on the Cardano blockchain using two strategies. This is step 5 in the Document Certification Flow. For Cardano best practices, see Section 9.4.

- **Single Anchor (Label 674):** One document hash per transaction for high-value documents
- **Merkle Batch (Label 675):** Multiple document hashes combined into a Merkle tree root for cost efficiency

Supports mock mode for development and Blockfrost API for preprod/mainnet. See Section 7.3 for network details.

PRISM Service

Manages Decentralised Identifiers (DIDs) supporting `did:prism` and `did:key` methods. Provides DID creation, resolution, update, and deactivation, as well as credential issuance and verification. Compliant with W3C DID Core 1.0.

Wallet Connector

Implements CIP-30 wallet integration supporting Nami, Eternl, Lace, and Flint wallets. Handles wallet registration, signature verification, and multi-network support (devnet, preprod, mainnet). See Section 9.4 for CIP compliance.

6.5 Verifiable Credentials

Standards Compliance

Fully compliant with the **W3C Verifiable Credentials Data Model 1.1** specification. See Section 9.3 for W3C standards adherence details.

Feature	Implementation
Proof Type	Ed25519Signature2020 (see Section 8.2)
Status Mechanism	StatusList2021 (bitstring-based revocation)
DID Methods	did:prism, did:key
Credential Subject	Document hash, type, issuer, certification metadata
Blockchain Anchor	Cardano transaction hash, block height, network (see Section 6.4)
Provenance	Linked provenance entries within credential (see Section 4.3)

Verification Checks

Seven verification checks are performed as part of the Verification Flow: structure validity, context validation, Ed25519 signature verification, status list check (revocation), expiration check, issuer DID resolution, and blockchain anchor verification.

6.6 Governance and Staking

DAO Governance Model

Token-weighted voting with tier multipliers. Participants register with their staked amount and receive vote power based on their tier. For the full governance model including proposal lifecycle and delegation, see Section 10.

Tier	Minimum Stake	Vote Multiplier
Basic	< 1,000 SECURA	1.0x
Staker	≥ 1,000 SECURA	1.0x
Validator	≥ 10,000 SECURA	1.5x
Partner	≥ 100,000 SECURA	2.0x

Default governance parameters: 50% approval threshold, 30% quorum, 7-day voting period, 24-hour timelock. See Section 10.2 for the proposal lifecycle.

6.7 Billing and Chargeable Events

Pricing Model

Revenue is generated through chargeable events in a document's lifecycle. Integration with Stripe handles fiat payments.

Event Type	ADA Price	USD Price	Typical Payer
Certification Signing	5 ADA	$2.00	Issuer / Certifier
Blockchain Anchor	2 ADA	$0.80	Issuer / Certifier
Verification	0.5 ADA	$0.20	Verifier
Verification Certificate	3 ADA	$1.20	Verifier
Rights Transfer	3 ADA	$1.20	Rights Holder
Share	Free	Free	—
Premium Timestamp	10 ADA	$4.00	Requesting Party
Evidence Export	1 ADA	$0.40	Requesting Party
API Access	0.1 ADA	$0.04	API Consumer (see SDK)

Volume Discounts

100+ events: 5% • 500+: 10% • 1,000+: 15% • 5,000+: 20% • 10,000+: 25%

6.8 Analytics and Reporting

The Analytics Service provides platform-wide metrics, real-time dashboards, and exportable reports. Accessible by System Administrators via the Admin Dashboard.

- **Overview metrics:** Total certifications, verifications, users, revenue
- **Certification metrics:** By type, by organisation, by time period
- **Governance metrics:** Active proposals, participation rates, vote power distribution (see Section 10)
- **Revenue metrics:** By event type, by organisation, volume discounts applied (see Section 6.7)
- **Real-time metrics:** Active users, pending operations, service health
- **Report generation:** Scheduled and on-demand reports with export to CSV/JSON

6.9 Privacy and Zero-Knowledge Proofs

The Midnight protocol integration enables selective disclosure of credential attributes without revealing the full credential. This supports the Subject's right to privacy and aligns with GDPR data minimisation principles.

Supported ZKP Circuits

Circuit	Purpose	Example
Ownership Proof	Prove ownership of a credential	"I hold a valid degree"
Age Verification	Prove age threshold without revealing date of birth	"I am over 18"
Credential Proof	Prove specific credential attributes	"My degree is in Computer Science"
Membership Proof	Prove membership in a group	"I am a member of Organisation X"
Range Proof	Prove a value falls within a range	"My salary is between $50K and $100K"

Database tables: `svc_zk_proofs`, `svc_commitments`, `svc_disclosure_audit` (see Appendix B).

7. Integration Points

Secuura integrates with multiple external platforms and protocols. This section covers each integration in detail. For the backend services that implement these integrations, see Section 5.2. For security considerations, see Section 8.

7.1 Microsoft 365 / Graph API

Overview

The M365 Integration Service connects Secuura to the Microsoft ecosystem via the Microsoft Graph API, enabling document synchronisation from SharePoint, identity federation via Entra ID, and Outlook integration. This is managed by System Administrators via the Admin Dashboard's Integration Hub.

Integration	Protocol	Endpoints
Microsoft Entra ID (Azure AD)	OAuth 2.0 / OIDC via MSAL	`/api/connections/*`
SharePoint Document Libraries	Microsoft Graph API v1.0	`/api/sites/*`, `/api/documents/*`
OneDrive File Access	Microsoft Graph API v1.0	`/api/documents/*`
Outlook Add-in SSO	OAuth 2.0 On-Behalf-Of	`/api/outlook/*`

Authentication Flow

The service uses MSAL (Microsoft Authentication Library) for OAuth 2.0 authentication. Tokens are managed with automatic refresh. Entra ID can also serve as an authentication method for platform access.

Environment Configuration

```
ENTRA_TENANT_ID=xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
ENTRA_CLIENT_ID=xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
ENTRA_CLIENT_SECRET=your-client-secret
ENTRA_REDIRECT_URI=https://app.secuura.com/auth/callback
```

Full environment reference in Appendix D.

7.2 Outlook Add-in

Architecture

A Microsoft Office Add-in built with React, Office.js, and Fluent UI v9 that integrates directly into the Outlook ribbon. It enables stakeholders to verify and certify email attachments without leaving Outlook. Hashing is performed client-side per the core design principles.

Capabilities

- **Verify Attachments:** Hash email attachments client-side (see Section 8.2) and check against the Secuura blockchain
- **Certify Documents:** Register uncertified attachments for certification
- **Quick Verify:** One-click ribbon button for instant batch verification
- **SSO:** Microsoft Entra ID single sign-on via On-Behalf-Of flow (see Section 7.1)

Deployment

Method	Scope	Description
Centralised Deployment	Organisation-wide	Deployed via M365 Admin Centre for all users
Sideload	Individual	Manual install via Outlook → Get Add-ins → Custom Add-ins
AppSource	Public	Published to Microsoft AppSource marketplace

Manifest URL: `https://[host]/outlook-addin/manifest.xml`

Dev server: `https://localhost:6104` (HTTPS required for Office Add-ins). See Appendix A for port details.

7.3 Cardano Blockchain

Integration Points

Service	Integration	Protocol	Reference
Anchoring	Document hash anchoring	Blockfrost API, metadata labels 674/675	Section 4.1
PRISM	DID management	Cardano PRISM, did:prism method	Section 9.3
Wallet Connector	Wallet authentication	CIP-30 (dApp connector), CIP-8 (message signing)	Section 6.3
NFT Certificate	Certificate minting	CIP-25 (NFT standard), CIP-30	Section 7.4
Staking	Token staking	Cardano staking delegation	Section 10
Governance	On-chain voting	Transaction metadata, datum hashing	Section 10.1

Network Support

- **Devnet:** Local Yaci DevKit for development (ports 6091–6093, see Appendix A)
- **Mock:** Mock Cardano responses for testing (port 6090)
- **Preprod:** Cardano pre-production testnet via Blockfrost
- **Mainnet:** Cardano mainnet via Blockfrost (production)

For Cardano Improvement Proposal (CIP) compliance, see Section 9.4.

7.4 IPFS and Decentralised Storage

The NFT Certificate Service supports multiple storage tiers for certificate metadata and assets:

Tier	Storage	Persistence	Use Case
On-Chain	Cardano transaction metadata	Permanent (blockchain)	Small metadata, high-value certificates
IPFS	Pinata / Infura / Web3.Storage	Permanent (pinned)	Standard certificates, documents
Arweave	Arweave permaweb	Permanent (incentivised)	Long-term archival
Centralised	Secuura CDN	Managed	Low-cost, convenience

The storage provider is configured via `IPFS_PROVIDER` (see Appendix D).

7.5 Stripe Payments

The Billing Service integrates with Stripe for fiat payment processing. Supports certification bundle purchases with credit tracking.

- **Checkout:** Stripe Checkout sessions for secure payment
- **Webhooks:** `/webhooks/stripe` handles payment confirmations, refunds, and disputes
- **Bundles:** Pre-purchased certification credit bundles (credits never expire)
- **Modes:** Test mode (development) and live mode (production)

Environment variables: `STRIPE_SECRET_KEY`, `STRIPE_WEBHOOK_SECRET` (see Appendix D).

7.6 KYC Providers

Identity verification via pluggable KYC providers, used to verify stakeholder identity before granting elevated privileges:

- **Mock:** Development testing with configurable results
- **Onfido:** Document verification and facial biometrics
- **Jumio:** AI-powered identity verification
- **Microsoft Entra ID:** Enterprise identity verification via `/api/kyc/microsoft/*` (see Section 7.1)

7.7 JavaScript SDK

The `@secuura/sdk` package provides a typed JavaScript/TypeScript client for integrating with the Secuura API. API access is a chargeable event.

Modules

- `CertificationsModule` — Create, verify, revoke certifications (see Section 4.1, Section 4.2)
- `GovernanceModule` — Proposals, voting, delegation (see Section 10)
- `PrivacyModule` — ZKP generation, selective disclosure (see Section 6.9)
- `TransfersModule` — Ownership transfers, delegations (see Section 4.4)
- `KycModule` — Identity verification workflows (see Section 7.6)
- `WebhooksModule` — Webhook subscription and handling

Build output: CommonJS + ESM + TypeScript declarations via tsup.

8. Security Architecture

Security is integral to every layer of the Secuura platform. This section details the controls in place. For compliance mapping to NIST CSF, see Section 9.2. For infrastructure security, see Section 5.4.

8.1 Authentication and Authorisation

For the full Authentication Service specification including supported methods (password, wallet, Entra ID, MFA), see Section 6.3.

Control	Implementation
Password Hashing	bcrypt with automatic salt generation
JWT Tokens	HS256, short-lived access tokens, httpOnly refresh cookies
Token Rotation	Automatic refresh with sliding window expiry
MFA	TOTP (RFC 6238) with encrypted backup codes
Wallet Auth	CIP-8 challenge/response with nonce-based replay prevention (see Section 7.3)
Session Management	Redis-backed with configurable TTL and forced revocation
RBAC	Role-based access control (SYSTEM_ADMIN, ORG_ADMIN, USER) mapped to stakeholder roles
Production Guard	JWT_SECRET validation fails if default value is used in production (see Appendix D)

8.2 Encryption and Hashing

Client-side hashing is a core design principle of the SVS. Documents never leave the stakeholder's device.

Purpose	Algorithm	Location
Document Hashing	SHA-256 (Web Crypto API) or BLAKE2b-256	Client-side (browser) — Core Principle
Password Hashing	bcrypt (adaptive cost factor)	Auth Service
API Key Hashing	SHA-256	Security Service
Token Hashing	SHA-256 (session tokens, refresh tokens)	Auth Service
VC Signing	Ed25519 (Ed25519Signature2020)	VC Issuer / Shared Package
TLS	TLSv1.2 / TLSv1.3 (ECDHE-ECDSA-AES128/256-GCM-SHA256/384)	Nginx Gateway
Database Auth	SCRAM-SHA-256	PostgreSQL (see Section 5.4)
Redis Auth	Password-based with requirepass	Redis

8.3 API Security

The API Gateway implements multiple security layers. For NIST CSF mapping of these controls, see Section 9.2.

Security Headers

Header	Value
X-Frame-Options	DENY (API Gateway) / SAMEORIGIN (Nginx)
X-Content-Type-Options	nosniff
X-XSS-Protection	1; mode=block
Referrer-Policy	strict-origin-when-cross-origin
Permissions-Policy	geolocation=(), microphone=(), camera=()
Strict-Transport-Security	max-age=63072000; includeSubDomains; preload (production)
Content-Security-Policy	default-src 'self'; script-src 'self' 'unsafe-inline'; connect-src 'self' https://*.blockfrost.io

Input Protection

- **XSS:** HTML entity encoding on all request body values
- **SQL Injection:** Parameterised queries, suspicious pattern detection
- **Path Traversal:** Pattern matching on URLs (`../` detection)
- **Command Injection:** `exec(` pattern detection in request bodies
- **CORS:** Explicit origin whitelist with credentials support

8.4 Infrastructure Security

For deployment architecture details, see Section 5.4. For the full port reference, see Appendix A.

Control	Implementation
Network Isolation	Docker bridge network; only ports 6881/6882 exposed externally (see Section 5.6)
Container Security	Non-root users, read-only filesystems where possible, resource limits
Secret Management	Environment variables, never committed to version control (see Appendix D)
Database Security	SCRAM-SHA-256 auth, dedicated user, init script-based schema
Redis Security	Password-protected, max memory limit (256MB), LRU eviction
Health Checks	All services implement `/health` endpoint with dependency status
Resource Limits	CPU and memory limits per container (e.g., API Gateway: 1 CPU, 512MB)
Logging	Structured JSON logging via Winston, no PII in logs (GDPR Art. 32)

9. Compliance and Standards Adherence

Secuura is designed to comply with international data protection, cybersecurity, identity, and blockchain standards. This section maps platform capabilities to specific regulatory and standards requirements. For the security controls that implement these standards, see Section 8.

9.1 GDPR Compliance

Secuura implements comprehensive GDPR compliance through the Originate Service's `/api/gdpr/*` endpoints. Subjects can exercise their data rights directly, and System Administrators can process requests via the Admin Dashboard.

GDPR Article	Requirement	Implementation
Art. 6	Lawful Basis for Processing	Consent records with purpose, version, source tracking. Consent can be withdrawn at any time.
Art. 7	Conditions for Consent	Explicit consent recording with IP address and user agent capture. Withdrawal endpoint available.
Art. 15	Right of Access	DSR type "access". Full user data export via `/api/gdpr/export/:userId`.
Art. 16	Right to Rectification	DSR type "rectification" with admin review workflow.
Art. 17	Right to Erasure	Cascading deletion across 9 data categories via `/api/gdpr/erasure/:userId`.
Art. 18	Right to Restriction	DSR type "restriction" with status tracking.
Art. 20	Right to Portability	JSON data export with download endpoint. Content-Disposition: attachment.
Art. 21	Right to Object	DSR type "objection" with 30-day processing deadline.
Art. 30	Records of Processing	Audit log table with user_id, action, resource_type, IP address, timestamp (see Appendix B).
Art. 32	Security of Processing	Encryption at rest, TLS in transit, access controls, pseudonymisation (see Section 8).
Art. 33/34	Breach Notification	Security event tracking via Security Service with severity classification.

Data Retention

Configurable retention policies per data type with automatic enforcement. Deletion methods include hard delete, soft delete, and anonymisation. All deletions are logged in `data_deletion_log` (see Appendix B).

Privacy by Design

The core design principle of client-side hashing means no original documents are stored, and zero-knowledge proofs enable selective disclosure without exposing full credentials.

9.2 NIST Cybersecurity Framework

Function	Category	Secuura Implementation	Reference
Identify	Asset Management	Service catalogue with health monitoring; database schema with data classifications	Section 5.2
Identify	Risk Assessment	Suspicious request detection; brute-force monitoring; security event classification	Section 6.1
Protect	Access Control	RBAC, JWT auth, MFA, wallet auth, session management	Section 8.1
Protect	Data Security	Client-side hashing, TLS 1.2/1.3, bcrypt, Ed25519, SCRAM-SHA-256	Section 8.2
Protect	Maintenance	Docker containerisation, version-controlled deployments, healthchecks	Section 5.4
Detect	Anomaly Detection	Suspicious pattern matching, rate limiting, brute-force detection	Section 8.3
Detect	Monitoring	Prometheus metrics, Grafana dashboards, structured logging	Section 5.4
Respond	Incident Response	Security event tracking, audit logs, IP-based blocking	Section 8.3
Recover	Recovery Planning	PostgreSQL persistent volumes, Redis AOF, Docker restart policies	Section 8.4

9.3 W3C Standards

Standard	Version	Implementation	Reference
Verifiable Credentials Data Model	1.1	Full implementation via VC Issuer Service and shared VC builder/verifier	Section 6.5
DID Core	1.0	did:prism and did:key methods via PRISM Service	Section 6.4
StatusList2021	Draft	Bitstring-based credential revocation status lists	Section 6.5
JSON-LD	1.1	VC context processing with standard credential contexts	—

9.4 Cardano and Crypto Best Practices

For the full Cardano integration architecture, see Section 7.3.

CIP	Title	Usage	Service
CIP-8	Message Signing	Wallet authentication challenge/response	Auth
CIP-25	NFT Metadata Standard	NFT certificate metadata structure	NFT Certificate
CIP-30	Cardano dApp Connector	Wallet connection (Nami, Eternl, Lace, Flint)	Wallet Connector
Label 674	Single Document Metadata	Individual document hash anchoring	Anchoring
Label 675	Batch Metadata	Merkle root anchoring for batch documents	Anchoring

Blockchain Best Practices

- Merkle tree batching reduces on-chain costs while maintaining individual verifiability
- Transaction confirmation polling with retry logic for reliable anchoring
- Network-agnostic configuration supporting devnet, preprod, and mainnet
- Mock mode for development eliminates blockchain dependency during testing

9.5 eIDAS and Timestamping

The Timestamping Service implements RFC-3161 timestamping via eIDAS-compliant Timestamp Authorities. This is step 7 in the Document Certification Flow.

- **TSA Provider:** D-Trust eIDAS TSA (configurable)
- **Protocol:** RFC 3161 Timestamp Protocol
- **Storage:** Timestamp tokens stored alongside document records in the database
- **Verification:** Timestamp tokens can be independently verified against the TSA
- **Legal Standing:** eIDAS-qualified timestamps provide legal non-repudiation across EU member states

10. Governance Contracts

This section details the decentralised governance model. For the underlying staking and tier system, see Section 6.6. For Cardano integration details, see Section 7.3. Governance metrics are available via the Analytics Service.

10.1 DAO Governance Model

Secuura implements decentralised governance through token-weighted voting, allowing SECURA token holders to participate in platform decisions. The staking tier system determines vote power.

Proposal Categories

Category	Description	Typical Use
`parameter_change`	Protocol parameter adjustments	Rate limit changes, pricing updates
`fee_adjustment`	Fee structure changes	Certification pricing, discount thresholds
`treasury_spend`	Treasury fund allocation	Development funding, marketing
`upgrade`	Protocol upgrades	New service deployments, breaking changes
`emergency`	Emergency actions	Security patches, incident response
`governance_change`	Governance rule changes	Voting period adjustments, quorum changes
`whitelist`	Whitelist management	Approved verifier/certifier lists
`general`	General proposals	Community initiatives

10.2 Proposal Lifecycle

1. **Draft** — Proposer creates proposal (requires minimum 1,000 SECURA staked per tier system)
2. **Active** — Proposal submitted, 7-day voting period begins
3. **Voting** — Participants cast votes (for/against/abstain) weighted by staked amount and tier multiplier
4. **Queued** — If passed (50% approval, 30% quorum), enters 24-hour timelock
5. **Executed** — After timelock, proposal actions are executed on-chain (see Section 7.3)

Proposals that fail to meet the approval threshold or quorum are automatically cancelled after the voting period ends (checked every 60 seconds by the governance scheduler). Database tables: `svc_proposals`, `svc_governance_participants` (see Appendix B).

10.3 Vote Delegation

Token holders can delegate their vote power to another participant, enabling representative governance.

- Delegation can be **universal** (all proposals) or **category-specific**
- Delegated power equals the delegator's `staked_amount × tier_multiplier` (see Section 6.6)
- Delegations can be revoked at any time
- Delegations can have an expiry date
- Supports DRep (Delegated Representative) and Constitutional Committee roles per Cardano best practices

The JavaScript SDK's `GovernanceModule` provides full programmatic access to delegation and voting.

11. Appendices

Appendix A — Complete Port Reference

All services communicate via the internal Docker bridge network. Only ports 6881 and 6882 are exposed externally. For network architecture, see Section 5.6.

Service	Internal Port	External (Host) Port	Gateway Route
Nginx Gateway	80	6882	Unified entry point
Website (direct)	80	6881	`/website/`
API Gateway	8080	—	`/api/`
Originate	4000	6000	`/originate/`
PRISM	4001	6001	—
Wallet Connector	4002	6002	—
Auth	4003	6003	—
Timestamping	4004	6004	—
Anchoring	4005	6005	—
Guardian	4006	6006	—
Queue	4007	6007	—
Security	4008	6008	—
Analytics	4010	6010	`/analytics/`
KYC	4011	6011	—
Midnight	4012	6012	—
M365 Integration	4013	6013	—
VC Issuer	4014	6014	—
Staking	4015	6015	—
Referral	4016	6016	—
Governance	4017	6017	—
NFT Certificate	4018	6018	—
Billing	4019	6019	—
Transfer	4020	6020	—
PostgreSQL	5432	6432	—
Redis	6379	6379	—
PGBouncer	6432	6433	—

Cardano Mock	80	6090	—
Cardano Devnet CLI	3001	6091	—
Cardano Submit API	8090	6092	—
Cardano Node	10000	6093	—
Grafana	3000	6300	—
Prometheus	9090	6301	—
Issuer Portal (dev)	—	6100	`/`
Verifier Portal (dev)	—	6101	`/verify/`
Admin Dashboard (dev)	—	6102	`/admin/`
Website (dev)	—	6103	`/website/`
Outlook Add-in (dev)	—	6104	—

Appendix B — Database Schema Reference

For the database architecture overview, see Section 5.5.

Core Tables

Table	Key Columns	Purpose	Section
`users`	id, email, password_hash, role, status, mfa_enabled	User accounts	6.3
`organizations`	id, name, slug, type, status, did	Organisation records	—
`documents`	id, content_hash, title, document_type, status	Document records (hash only)	6.2
`certification_events`	id, document_id, event_type, event_hash, previous_event_id	Provenance chain	4.3
`blockchain_anchors`	id, document_id, anchor_type, content_hash, transaction_hash	Blockchain records	6.4
`verifiable_credentials`	credential_id, issuer_did, holder_did, proof, status	W3C VCs	6.5
`consent_records`	user_id, purpose, version, given_at, withdrawn_at	GDPR consent	9.1
`data_subject_requests`	type, user_id, status, deadline, completed_at	GDPR DSRs	9.1
`audit_logs`	user_id, action, resource_type, ip_address, success	Audit trail	9.1

Appendix C — API Endpoint Catalogue

For the API Gateway configuration and security, see Section 6.1 and Section 8.3.

Service	Base Path	Key Endpoints	Section
Auth	`/api/auth`	login, register, refresh, logout, verify-email, reset-password, wallet/*, mfa/*	6.3
Certifications	`/api/certifications`	create, verify, revoke, share, recertify, lineage, evidence-bundle	4.1
Documents	`/api/documents`	create, sign, anchor, qr, list, detail	6.2
Verification	`/api/verification`	verify-hash, verify-integrity, check-status	4.2
GDPR	`/api/gdpr`	consent, consent/withdraw, dsr, export, erasure, retention	9.1
Anchoring	`/api/anchors`	create, batch, verify, queue	6.4
DIDs	`/api/did`	create, resolve, update, deactivate	9.3
Credentials	`/api/credentials`	issue, verify, revoke, list	6.5
Governance	`/api/governance`	proposals/*, vote, delegate, participants/*, config, stats	10
Staking	`/api/stake`	stake, unstake, rewards/*, tiers/*	6.6
NFT	`/api/nft`	record, estimate, certificates/*, tiers, privacy-options	7.4
M365	`/api/connections`	connect, callback, refresh, sites/*, documents/*, outlook/*	7.1
Analytics	`/api/analytics`	overview, certifications, verifications, governance, revenue, realtime	6.8
Billing	`/billing`	bundles/*, checkout/*, customers/*	6.7
Admin	`/api/admin`	document-types, workflows, integrations, settings, users, health	—

Appendix D — Environment Variable Reference

All sensitive configuration is managed via environment variables. See Section 8.4 for secret management practices.

Variable	Service	Description	Reference
`JWT_SECRET`	All	JWT signing secret (must be changed in production)	8.1
`DATABASE_URL`	All	PostgreSQL connection string	5.5
`REDIS_URL`	All	Redis connection string	5.4
`CORS_ORIGINS`	Gateway	Comma-separated allowed origins	8.3
`BLOCKFROST_API_KEY`	Anchoring, NFT	Blockfrost API project ID	7.3
`CARDANO_NETWORK`	Blockchain	devnet, preprod, or mainnet	7.3
`ENTRA_TENANT_ID`	M365	Microsoft Entra ID tenant	7.1
`ENTRA_CLIENT_ID`	M365	Azure app registration client ID	7.1
`ENTRA_CLIENT_SECRET`	M365	Azure app registration client secret	7.1
`STRIPE_SECRET_KEY`	Billing	Stripe API secret key	7.5
`STRIPE_WEBHOOK_SECRET`	Billing	Stripe webhook signing secret	7.5
`IPFS_PROVIDER`	NFT	IPFS provider (pinata, infura, web3storage, mock)	7.4
`RATE_LIMIT_MAX_REQUESTS`	Gateway	Maximum API requests per minute	6.1

Appendix E — Glossary

Term	Definition	Reference
SVS	Secuura Verification System — the overarching platform model	Section 2
Certifier	Entity that attests to the authenticity and integrity of information	Section 3.2
Verifier	Entity that checks a Certifier has certified information	Section 3.5
Rights Holder	Entity with ownership or control rights over information	Section 3.4
DID	Decentralised Identifier — a W3C standard for self-sovereign identity	Section 9.3
VC	Verifiable Credential — a W3C standard for tamper-evident credentials	Section 6.5
CIP	Cardano Improvement Proposal — community standards for Cardano	Section 9.4
ZKP	Zero-Knowledge Proof — prove a statement without revealing the underlying data	Section 6.9
DAO	Decentralised Autonomous Organisation — governance via token-weighted voting	Section 10
eIDAS	EU regulation for electronic identification, authentication, and trust services	Section 9.5
MSAL	Microsoft Authentication Library — OAuth 2.0 client for Microsoft identity	Section 7.1
Merkle Tree	Hash tree structure used for efficient batch verification of multiple documents	Section 6.4
TSA	Timestamp Authority — provides RFC-3161 trusted timestamps	Section 9.5

End of Document

Secuura Verification System — Technical Specifications and Platform Overview

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